

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 HSC ASSESSMENT TASK 2

MARCH 2005

MATHEMATICS

Extension 1

Time Allowed: 70 minutes

Instructions:

- Attempt all questions
- Start each question on a new page
- Show all necessary working
- The marks for each question are indicated next to the question
- Marks may be deducted for careless or badly arranged work
- Approved calculators may be used
- Marks indicated are a guide only and may be varied if necessary

Name: _____ Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Total

QUESTION 1

- a) Find $\int (3x + 5)^6 dx$ (1)
- b) (i) Differentiate $(x^3 - 1)^6$ (1)
(ii) Hence find $\int x^2 (x^3 - 1)^5 dx$ (2)
- c) Show why the curve $y = -x^3 + 6x^2$ has a non-horizontal point of inflexion when $x = 2$ (2)
- d) Find the equation of the tangent in general form, to the curve $y = x^3 - 6x^2 + 3x + 2$ at the point where $\frac{d^2y}{dx^2} = 0$ (4)

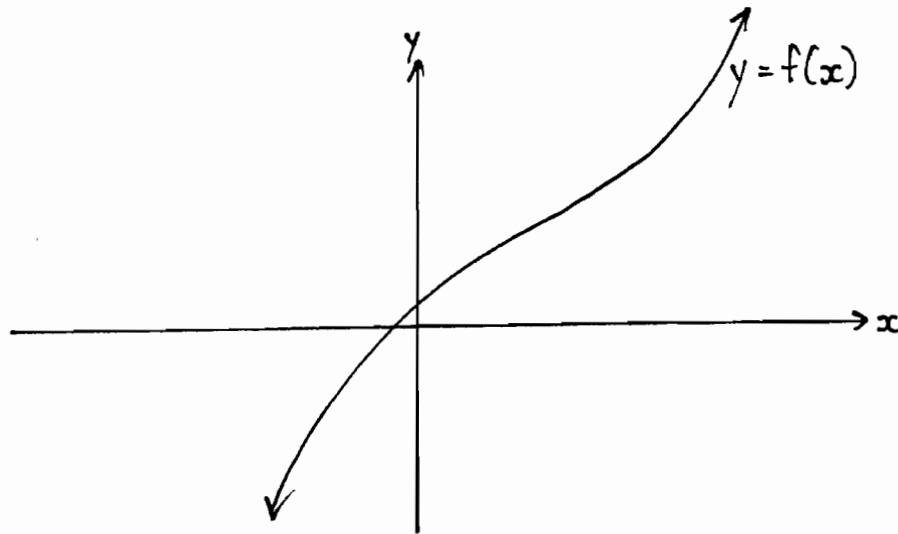
QUESTION 2 (Start a new page)

- a) For the curve $f(x) = \frac{x}{x^2 - 4}$
- (i) Show that it is an odd function (1)
- (ii) Show that $f(x)$ has no stationary points (2)
- (iii) Find the equations of the vertical asymptotes (1)
- (iv) Hence sketch the curve (2)

QUESTION 2 (cont)

b) Copy the graph of $y = f(x)$ below onto your answer sheet.

Sketch a graph of $y = f'(x)$ on the same number plane, given $f'(0) = 4$ (2)



c) For what values of x is $f(x) = x^3 - 3x^2$ concave down? (2)

QUESTION 3 (Start a new page)

a) In the diagram, $PQRS$ is a rectangle with $PQ = 40\text{cm}$ and $SP = 10\text{cm}$.

The shaded portions are cut away, leaving the parallelogram $KLMN$

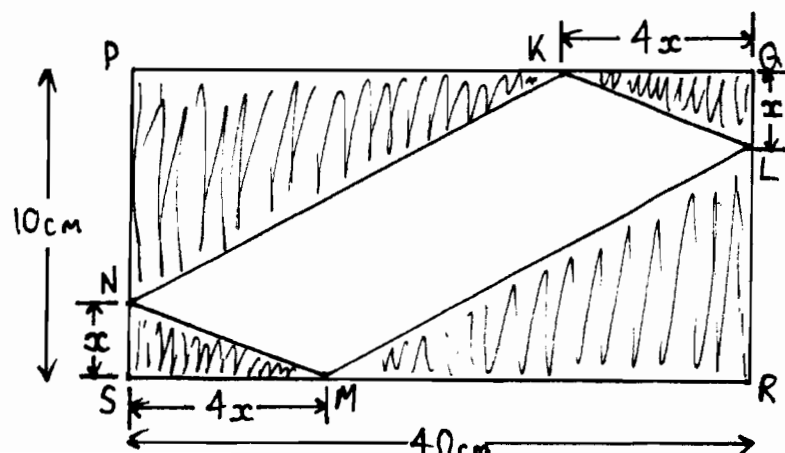
$QL = SN = x\text{cm}$ and $QK = SM = 4x\text{cm}$

(i) Show that the area of the parallelogram $KLMN$ is given by (2)

$$A = 80x - 8x^2$$

(ii) Find the allowable values of x (1)

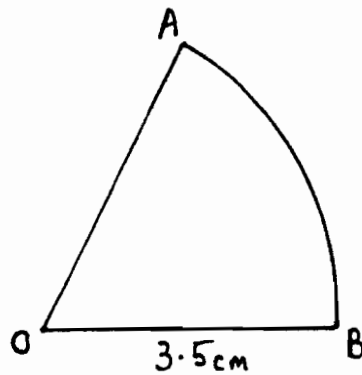
(iii) Find the value of x for which A is a maximum (2)



Question 3 cont

b) A sector AOB of a circle has a radius of 3.5cm . Its perimeter is 9.5cm

- (i) Find the size of $\angle AOB$ in radians correct to one decimal place (2)
- (ii) Find the length of the arc AB correct to one decimal place (1)
- (iii) Find the area of the sector AOB correct to one decimal place (2)

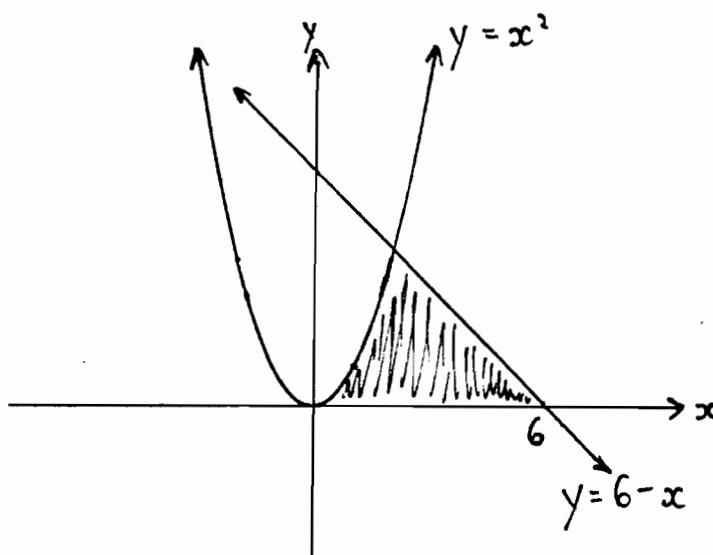


QUESTION 4 (Start a new page)

- a) Find the equation of $\frac{dy}{dx}$ given

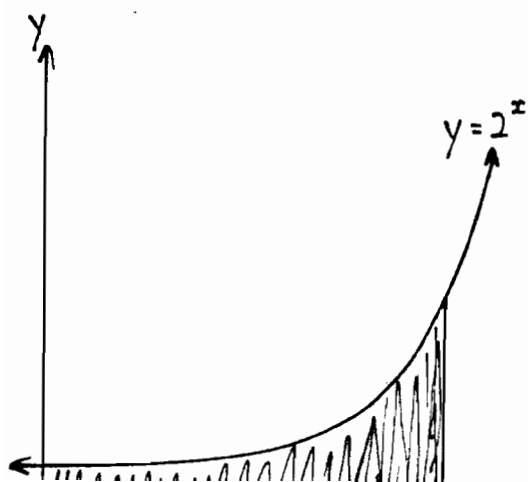
$$\frac{d^2 y}{dx^2} = 2x \text{ and when } x = 1, \frac{dy}{dx} = 2. \quad (2)$$

- b) (i) Find the x value of the point of intersection in the 1st quadrant of the 2 functions below (2)



- (ii) Hence find the exact value of the shaded area (3)

- c) Use the function values in the table below to estimate the shaded area using the Trapezoidal Rule (leave your answer in exact form)

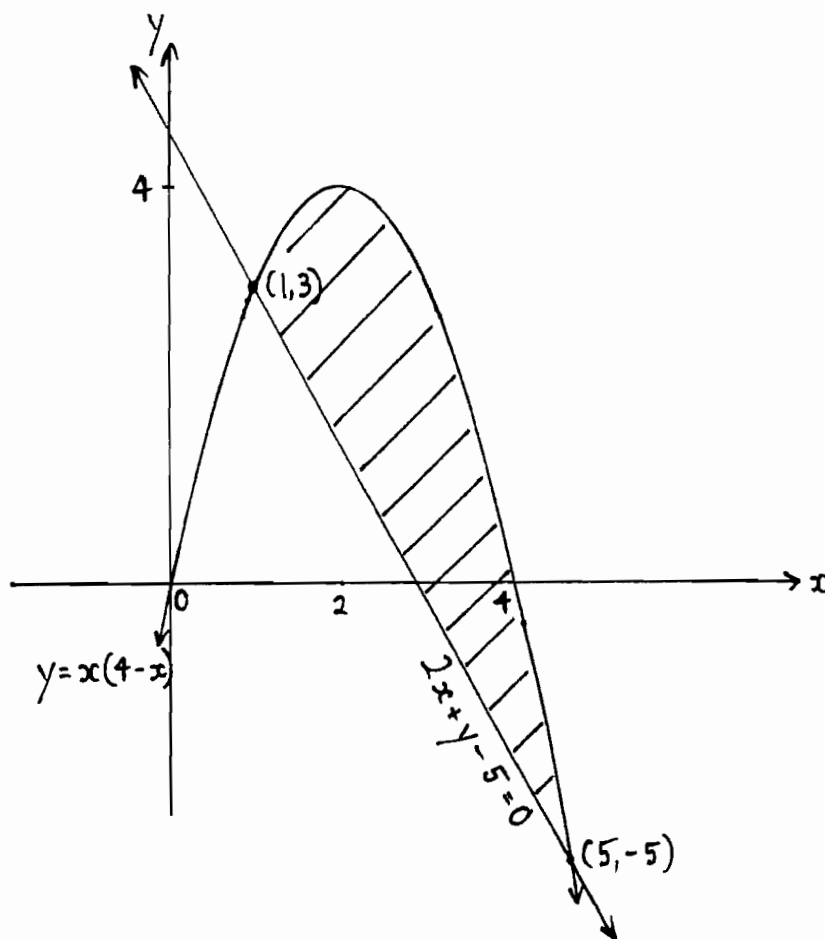


x	0	0.5	1	1.5	2
y	1	$\sqrt{2}$	2	$2\sqrt{2}$	4

QUESTION 5 (Start a new page)

a) Evaluate $\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{x^2}$ (2)

- b) On the graph below, the functions $2x + y - 5 = 0$ and $y = x(4 - x)$ intersect at the points indicated. Find the shaded area. (4)



c) Find $\int_0^3 \frac{x}{\sqrt{1+x}} dx$ using the substitution $x = u^2 - 1$ where $u > 0$ (4)

QUESTION 6 (Start a new page)

- a) The curve $y = \frac{4}{x}$ is rotated around the x – axis between $x = 1$ and $x = 3$ to form a solid.

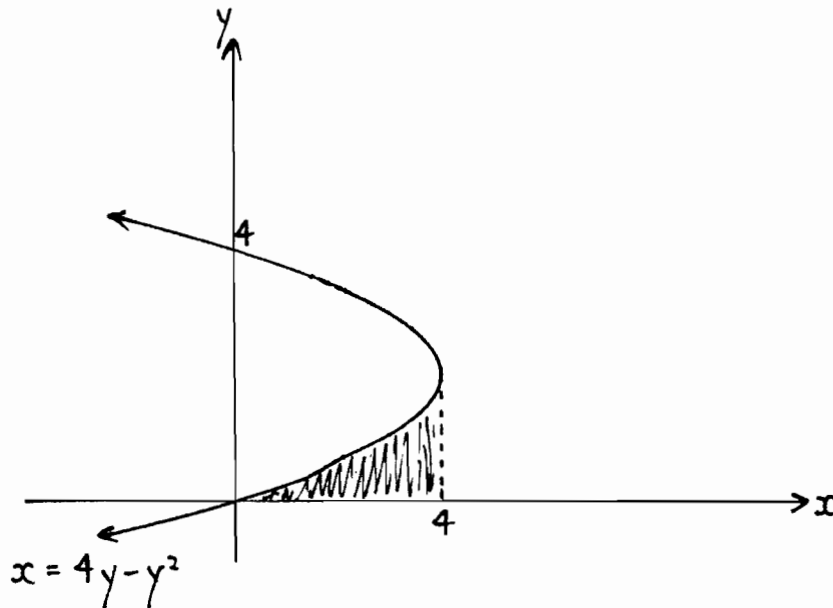
(i) Write down the integral which gives this volume (1)

(ii) Using your answer to part (i), complete a table with 5 function values for $x = 1, 1.5, 2, 2.5, 3$ (1)

(iii) Use Simpsons Rule to estimate the volume correct to one decimal place (2)

- b) Sketch the curve $y = 2 \sin(2x + \pi)$ for $-\pi \leq x \leq \pi$ (3)

- c) Find the exact value of the shaded area below (3)



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Question 1

$$\int (3x+5)^6 dx$$

$$= \frac{(3x+5)^7}{7 \times 3}$$

$$= \frac{(3x+5)^7}{21} + C \quad \textcircled{1} \text{ give if they forget } +C$$

$$\text{c) } \frac{d}{dx} (x^3-1)^6$$

$$= 6(x^3-1)^5 \times 3x^2$$

$$= 18x^2(x^3-1)^5 \quad \textcircled{1}$$

$$\text{c) } \int x^2(x^3-1)^5 dx$$

$$\frac{d}{dx} (x^3-1)^6 = 18x^2(x^3-1)^5$$

$$\frac{1}{18} \frac{d}{dx} (x^3-1)^6 = x^2(x^3-1)^5 \quad \textcircled{1}$$

$$\int \frac{1}{18} \frac{d}{dx} (x^3-1)^6 dx = \int x^2(x^3-1)^5 dx$$

$$= \frac{1}{18} (x^3-1)^6 + C \quad \textcircled{1}$$

$$y = -x^3 + 6x^2$$

$$y' = -3x^2 + 12x$$

$$y'' = -6x + 12 = 0 \text{ for a pt. of inflexion}$$

$$6x = 12$$

$$x = 2 \quad \textcircled{1}$$

Since $y' \neq 0$ at $x=2$ it must be a non-horiz. pt. of inflexion $\textcircled{1}$

OR

x	1	2	3
y'	+	0	+

x	1	2	3
y''	-	0	+

$$y = x^3 - 6x^2 + 3x + 2$$

$$y' = 3x^2 - 12x + 3$$

$$y'' = 6x - 12 = 0$$

$$\Rightarrow x = 2 \therefore y = -8 \quad \textcircled{1}$$

$$\text{At } x = 2, y' = m = -9 \quad \textcircled{1}$$

$$y - y_1 = m(x - x_1) \quad \textcircled{1}$$

$$y + 8 = -9x + 18$$

$$9x + y - 10 = 0 \quad \textcircled{1}$$

Teacher's Name: _____ Student's Name/Nº: _____

Question 2

$$\text{a) i) Odd if } f(-x) = -f(x)$$

$$\frac{-x}{(-x)^2 - 4} = - \frac{x}{x^2 - 4}$$

$$\frac{-x}{x^2 - 4} = - \frac{x}{x^2 - 4} \quad \textcircled{1}$$

$$\text{c) Stat. pts where } y' = 0$$

$$y' = \frac{x^2 - 4}{(x^2 - 4)^2} - x \cdot 2x$$

$$= \frac{x^2 - 4}{(x^2 - 4)^2} = 0 \quad \textcircled{1}$$

$$\text{if } -x^2 - 4 = 0$$

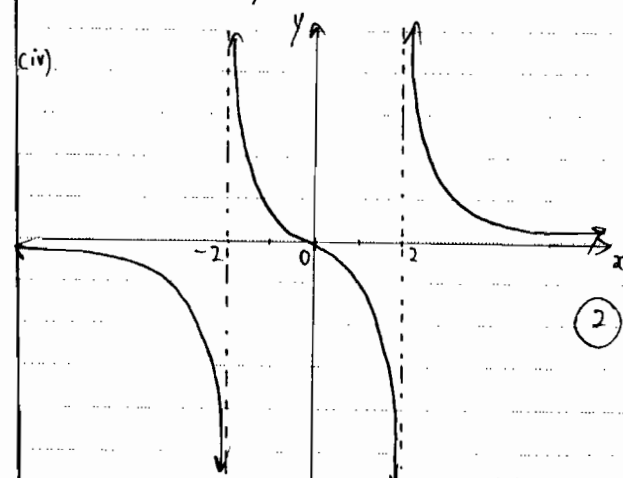
$$x^2 = -4 \quad \textcircled{1}$$

No sol'n.

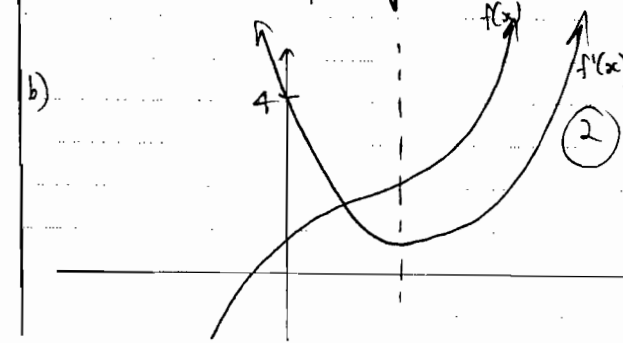
$$\text{c) ii) Vertical Asymptotes}$$

$$\text{where } x^2 - 4 = 0$$

$$\text{ie: } x = 2, x = -2 \quad \textcircled{1}$$



- 1 if $y=0$ not an asymptote
- 1 if outside branches OK but between $x = \pm 2$ wrong



- 1 if parabola vertex not in line with inflexion
- 1 if parabola touches or cuts x axis

Teacher's Name:

Student's Name/N°:

c) $f(x) = x^3 - 3x^2$ is concave down where
 $f''(x) < 0$
 $f'(x) = 3x^2 - 6x$
 $f''(x) = 6x - 6$ ①
 $6x - 6 < 0$
 $x < 1$ ①

Question 3

a) (i) $A = 40 \times 10 - 2 \times \frac{1}{2} \times 4x \times x - 2 \times \frac{1}{2} (40 - 4x) \times (10 - x)$ ①
 $= 400 - 4x^2 - (40 - 4x)(10 - x)$
 $= 400 - 4x^2 - (400 - 40x - 40x + 4x^2)$
 $= 400 - 4x^2 - 400 + 40x + 40x - 4x^2$
 $= 80x - 8x^2$ as required ①

(ii) $0 < x < 10$ ①
 $x < 10$ OK for ①

(iii) $\frac{dA}{dx} = 80 - 16x = 0$ for a maximum
 $80 = 16x$
 $x = 5$ ①

x	4	5	6
$\frac{dA}{dx}$	+	0	-

$\therefore x = 5$ gives a maximum ①

b) (i) $P = 9.5 = 3.5 \times 2 + r \times \theta$ ①
 $9.5 = 7 + 3.5 \times \theta$
 $2.5 = 3.5 \times \theta$
 $\theta = 0.7$ radians ①

(ii) $l = r \times \theta$
 $l = 3.5 \times 0.7$
 $= 2.5$ cm ①

(iii) $A = \frac{1}{2} r^2 \theta$ ①
 $= \frac{1}{2} \times 3.5^2 \times 0.7$
 $= 4.3$ cm² ①

Deduct 1 mark once
 only if answers not
 given correct to one d.p.

Teacher's Name:

Student's Name/N°:

Question 4

a) $\frac{dy}{dx} = 2x$
 $\frac{dy}{dx} = x^2 + c$ ①
 When $x = 1$, $\frac{dy}{dx} = 2$
 $2 = 1^2 + c$
 $\therefore c = 1$
 $\therefore \frac{dy}{dx} = x^2 + 1$ ①

b) (i) Solve $\begin{cases} y = x^2 \\ y = 6 - x \end{cases}$ simulta
 $\therefore x^2 = 6 - x$
 $x^2 + x - 6 = 0$ ①
 $(x - 2)(x + 3) = 0$
 $x = 2$ in 1st quadrant

(ii) $A = \int_0^2 x^2 dx + \int_2^6 6 - x dx$ ①
 $= \left[\frac{x^3}{3} \right]_0^2 + \left[6x - \frac{x^2}{2} \right]_2^6$ ①
 $= \frac{8}{3} + 36 - \frac{36}{2} - \left(12 - \frac{4}{2} \right)$
 $= \frac{8}{3} + 18 - 10$
 $= 10\frac{2}{3}$ units² ①

c) $A \div \frac{h}{2} (y_0 + y_n + 2 \sum y_{\text{others}})$ ①
 $= \frac{0.5}{2} (1 + 4 + 2(\sqrt{2} + 1 + 2\sqrt{2}))$
 $= \frac{1}{4} (7 + 6\sqrt{2})$ ①

Question 5

a) $\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{1 - 2\sin^2 x - 1}{x^2}$ ①
 $= -2 \lim_{x \rightarrow 0} \frac{\sin^2 x}{x^2}$
 $= -2 \times 1^2$
 $= -2$ ①

b) $A = \int_1^5 x(4 - x) - (-2x + 5) dx$
 $= \int_1^5 4x - x^2 + 2x - 5 dx$
 $= \int_1^5 6x - x^2 - 5 dx$
 $= \left[3x^2 - \frac{x^3}{3} - 5x \right]_1^5$
 $= 75 - \frac{125}{3} - 25 - \left(3 - \frac{1}{3} - 5 \right)$
 $= 10\frac{2}{3}$ ①

Teacher's Name: _____

Student's Name/N^o: _____

$$\int_0^1 \frac{x}{\sqrt{1+x}} dx$$

$$x = u^2 - 1 \quad \therefore 1+x = u^2$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

When $x=0$, $u=1$ ($u>0$)
 When $x=3$, $u=2$ ($u>0$)

$$\int_1^2 \frac{u^2-1}{u} 2u du \quad \textcircled{1}$$

$$2 \int_1^2 (u^2-1) du$$

$$2 \left[\frac{u^3}{3} - u \right]_1^2 \quad \textcircled{1}$$

$$= 2 \left(\frac{8}{3} - 2 - \left(\frac{1}{3} - 1 \right) \right)$$

$$= \frac{8}{3} \text{ units}^2 \quad \textcircled{1}$$

Question 6

$$(i) V = \pi \int y^2 dx$$

$$= \pi \int_1^3 \left(\frac{4}{x} \right)^2 dx \quad \textcircled{1}$$

$$= \pi \int_1^3 \frac{16}{x^2} dx \quad \text{or}$$

(ii)

x	1	1.5	2	2.5	3
y	16	7.11	4	2.56	16/9

y_0 y_1 y_2 y_3 y_4 $\textcircled{1}$

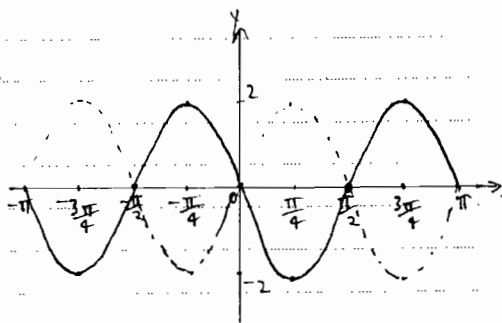
$$(iii) V = \pi \times \frac{h}{3} [y_0 + y_n + 4(y_{\text{odds}}) + 2(y_{\text{even}})] \quad \text{OR } \textcircled{1}$$

$$= \pi \times \frac{0.5}{3} \left[16 + \frac{16}{9} + 4(7.11 + 2.56) + 2 \times 4 \right] \quad \checkmark$$

$$= 33.8 \quad \textcircled{1}$$

$$b) y = 2 \sin(2x + \pi)$$

Amplitude 2

Period $\frac{2\pi}{2} = \pi$ Phase shift $\frac{\pi}{2}$ left-1 for
each
error

Teacher's Name: _____

Student's Name/N^o: _____

$$c) A = \int x dy$$

When $x=4$,
 $4 = 4y - y^2$
 $y^2 - 4y + 4 = 0$
 $(y-2)^2 = 0$
 $y = 2 \quad \textcircled{1}$

$$\text{Area} = \text{Rectangle} - \int_0^2 x dy$$

$$= 2 \times 4 - \int_0^2 (4y - y^2) dy \quad \textcircled{1}$$

$$= 8 - \left[2y^2 - \frac{y^3}{3} \right]_0^2$$

$$= 8 - \left(8 - \frac{8}{3} - 0 \right)$$

$$= \frac{8}{3} \text{ or } 2\frac{2}{3} \text{ units}^2 \quad \textcircled{1}$$